cally predictable, the technique to create the audio beam is simple; modulate the amplitude to get the hybrid wave, then calculate what the Becktay's Equation does to this signal, and do the exact opposite. In other words, distort it, before Mother Nature does it. Finally, pass this wave through air, and what you get is the original sound wave component whose volume, this time, is exponentially related to the volume of the ultrasound beam, and a distorted component, whose volume now varies directly as the ultrasound wave. Once you crank up the volume of the ultrasound, you can get as high as 80 decibels with just 5 per cent distortion—not the best in the world yet, but a far cry from the 50 per cent or so that the Japanese teams faced. The best speakers in the world can currently get distortion down to below 1 per cent and research is getting the targeted audio beam very close to this point.

Since you can control and focus the ultrasound beam just like a flashlight, you can direct it such that you would hear the sound only if you were in the path of the beam. This is called directed sound. You could also bounce the beam off a reflecting surface, so that people in the path of the audio reflection can hear the sound. This is known as projected audio. In short, unlike ordinary speakers, you will hear the sound only if you disrupt the sound beam, whether you stand in its path or in the path of a reflection from an acoustic mirroring surface. If you step away from the path of the sound, you will hear nothing. In either case, the sound's source is not the physical device you see, but the invisible ultrasound beam that generates it.

This is where you are going to experience audio spotlighting

Personalised messaging: Using targeted sound, you could message people in high activity areas such as construction sites, and factory floors.

Acoustic assault rifles: You could employ sound to fire focussed, high-energy acoustic bullets that could be used as non-lethal weapons to deter crowds

Realistic movies: With targeted sound, movies could become more realistic, with the sound moving, along with its source on the screen.

Discreet announcements: In museums and exhibitions, using audio spotlighting techniques to discreetly inform people, without raising ambient sound levels

Consumer electronics: You could use focussed sound in televisions, and home theatre systems, to give realistic sound effects.

INFOGRAPH: Parag Joshi

U2 and us...

U2 front man Bono, is reportedly interested in systems such as the Audio Spotlight—and U2 is now working with the Holosonic Labs to create Audio Spotlights for rock concerts, where they are looking at giving users the kind of wild sound shows that are currently possible with light.

The sound ray is coming

The targeted or directed audio technology is going to tap a huge commercial market in entertainment and in consumer electronics, and the technology developers are scrambling to tap into that market. Joseph Pompei's Holosonic Research Labs developed the Audio Spotlight that is made of a sound processor, an amplifier and the transducer. Aim the transducer anywhere, and direct and project a three-degree wide sound beam that is audible even at 100 metres. The American Technology Corporation developed the HyperSonic Sound-based Directed Audio Sound System. Both use ultrasound based solutions to beam sound into a focused beam.

With something as radical as this coming into mainstream sound reproduction, analysts claim that this is possibly the most dramatic change in the way we perceive sound since the invention of the coil loudspeaker. Sound as we know it, is a very three-dimensional sensation. But if you strip out the perception of depth and direction, for example whilst watching a movie on TV, we tend to associate the sound source as being very fixed, and hence unreal. This is where three-dimensional sound beaming comes in. The technology that the Holosonic Research Labs and the American Technology Corporation are lining up may seem to be a novelty of sorts, but a wide range of applications are being targeted at it.

The first lab rats are going to be those attending exhibitions and trade shows. With targeted announcements at crowded shows, audio spotlighting technology may possibly be the best way for exhibitors to draw people to their stalls and demonstrate their product to potential customers. Supermarkets and retail stores may not be very far behind either—stand in this aisle that features detergents and hear the sales pitch for a new washing detergent, stand in the other and find out about the latest meat bargains. Retail displays and kiosks are possibly going to be the first to get wired up. Soon after this, spotlighting technology can also be built into vending machines, which might entice you to stop by for a refreshing cold drink as you pass by. There is an even bigger market for personalised sound systems in entertainment and consumer electronics. You could be wired up for spotlighting in the future with your home theatre audio system, and could experience sound as it was originally heard, possessing direction and movement, and this will be tuned to your favourite couch-potato position. If you just have to watch that show that everybody detests, then you could use a personalised spotlight that turns off the audio for everybody, except you, since you will be sitting right in the middle of it. Movie theatres could also jump onto the bandwagon, with live moving voices coming in from all over the screen, movies would be truer